

Retail Pulse AI — Synthetic Retail Dataset

Methodology, validation evidence, and known limitations

Prepared for management review · Dataset version: seed 20240101 (deterministic) · Every figure in this document is computed directly from the delivered CSV files, not typed by hand.

2,50,000 TRANSACTION LINES	7,80,000 FORECASTING PANEL ROWS	INR 21.02 Cr REVENUE MODELLED	8,000 CUSTOMERS	1,200 PRODUCTS
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1. What this is, in one paragraph

A purpose-built **synthetic** retail dataset for an Indian grocery chain, covering **2024-01-01 to 2025-12-31** (731 days). It is generated by a Python program — not collected from real customers — and models the behaviour we actually need to analyse: which products sell, in which store, in which city, at what price, under which promotion, and what happens to sales when stock runs out.

It ships as two files. The **transaction file** holds 2,50,000 individual line-items across 60,538 shopping transactions — the grain you use for customer insight, basket analysis and revenue reporting. The **forecasting panel** holds 7,80,000 rows at store × product × week grain, covering 104 complete weeks from **2024-01-01 to 2025-12-22**, across 50 stores and 150 products. It is already aggregated and deliberately dense: 77.9% of rows are genuine zero-demand weeks, which is what a real demand-forecasting input needs and what a naive aggregate would destroy.

Why synthetic rather than real retail data

Published retail datasets are usually wholesale led, cover 2009–2011, and carry no store, city, promotion or inventory fields — so they cannot answer Indian retail questions. Synthetic data also gives us something real data never can: the **true answer is known**, so model accuracy is measurable rather than assumed.

2. How it was built

- Scope was fixed first.** 50 stores across 12 Indian cities, 1,200 products in 10 categories, 8,000 customers, and the period 2024-01-01–2025-12-31. The row budget was capped at exactly 2,50,000 line-items.
- Structure was modelled, not faked.** Each product belongs to a category with its own pack sizes and price band. Each store carries an assortment; each customer has a loyalty tier and a segment. Every product is guaranteed to be stocked somewhere and to actually sell, so no product or customer is a dead record.
- Demand was shaped deliberately.** Weekly seasonality, Indian festival peaks (Diwali, New Year, Republic Day), weekday and weekend effects, category-specific seasonality, and a realistic long-tail where a small slice of the catalogue drives most revenue. Baskets are multi-item and vary by customer type, including bulk shoppers.
- Inventory was simulated causally.** Stock levels drive sales, not the reverse. When a shelf empties, a stock-out line is recorded with zero quantity — that is lost demand, not a data error. 7,500 such lines (3%) exist, and the forecasting panel carries the 6,959 affected store-weeks so demand models can learn from them.
- Every row was held to arithmetic rules.** Revenue must equal quantity multiplied by price less discount; a discount may only exist on a genuinely promoted line; zero quantity implies zero stock remaining; no quantity or stock may be negative. These are enforced, not assumed.
- Output is reproducible.** A fixed random seed means re-running the generator reproduces both files byte-for-byte. Any number in this document can be re-derived from the delivered CSVs.

3. Why the numbers look realistic

Rather than trusting the generator, its output was **calibrated against a real, published retail dataset** — the UCI Online Retail II dataset, 1,067,371 transaction lines from the UK, 2009–2011. Real data was used as a structural benchmark, not copied, because it does not match the business context.

Structural property	Real reference data	Our dataset	Read
SKUs needed for 50% of revenue	5.2%	4.8%	Long tail present and correctly shaped
SKUs needed for 80% of revenue	19.6%	19.4%	Concentration closely matched
Line-items per SKU	201	208	Product velocity distribution matches
Transactions per customer (mean)	7.6	7.6	Repeat-purchase behaviour matches

Price levels were deliberately *not* copied: the reference data is UK giftware at a £2.10 median, whereas this is Indian grocery at a INR 217 median line value. The reference dataset also mixes in wholesale buyers, so its headline basket size of 20 items is not representative of a consumer shop — basket size here is 3 at the median, with 14.6% of baskets being a single item, which matches consumer retail behaviour.

4. Evidence of quality

Every delivered file was checked by an automated test suite. These checks run on demand and fail loudly rather than passing silently.

Check	Result	Meaning
Specification conformance	54 / 54	Row count, column order, ranges, formula integrity, seasonality, calibration targets
Missing values	0	Across all 27 columns, including several spellings of null
Duplicate rows	0	Including the natural key of each file, so no grouping double-counts
Date validity	0 bad	731 consecutive days, no gaps, no invalid dates, derived columns agree
Panel ↔ sales consistency	exact	Panel reconciles to the sales file to the rupee, with a documented scope difference
Validator self-test	8 / 8	Confirms the validator actually catches corrupted data

A note on the validator

A data check that always passes is worthless. So the validator is itself tested against deliberately corrupted data: each of the eight core rules is broken in turn, and the validator must catch every one. It catches all eight.

5. What the dataset can be used for

- **Demand forecasting** — the dense weekly panel is a ready training input.
- **Customer segmentation** — 8,000 customers with 87.3% repeat rate and 85.4% multi-item baskets.
- **Promotion effectiveness** — 21,004 promoted lines across 6 campaigns, with discounts recorded per line.
- **Basket and affinity analysis** — transaction identifiers group lines into true baskets; no product repeats within a basket.
- **Revenue reporting** — daily, weekly, or per city, store or category.

Channel	Lines	Share
Store	2,07,174	82.9%
Mobile_App	31,412	12.6%
Web	11,414	4.6%

6. Known limitations — please read before use

These are genuine constraints, stated plainly. Any one of them is a reason to treat findings as directional rather than exact.

Limitation	Impact	Workaround
No returns ABSENT	Real retail is 2–4% returns. Here quantity is always zero or positive, so return-driven revenue reversal cannot be modelled.	Model returns as a separate rate applied on top.
No cost or margin ABSENT	Only price and discount are present, so profitability, margin and contribution analysis cannot be done.	Join a cost table from the product master.
No customer demographics LIMITED	Only loyalty tier and segment exist — no age, gender or income.	Use segment as the proxy, or join an external profile.
Panel covers 150 of 1,200 products SUBSET	A forecast cannot be produced for the other 1,050 products from the panel alone. The panel also stops at 2025-12-22 so that every week it contains is complete.	Extend the panel by re-running the generator with a larger product list.
Final week is incomplete TIMING	Data ends 2025-12-31, so the week starting 2025-12-29 holds only 3 of its 7 days and shows about 37% of a normal week.	The bundled weekly roll-up drops it automatically.
Synthetic, not observed NATURE	Behaviour is modelled, not measured. It will not capture a real promotion that failed, or a supply shock the model does not know about.	Validate findings against live data before any commercial commitment.

Recommended use

This dataset is best suited to **development, model training, pipeline testing and demonstrating approach** — situations where the right answer is known and accuracy can be measured. It should **not** be the sole basis of a pricing, stocking or investment decision. Real transaction data remains the reference for anything commercially binding.

7. Likely questions, and how to answer them

For the conversation itself. The left column is the question; the right is a short, plain-language answer you can give.

"Is this real customer data?"	No — it is generated by a program, not collected from customers. It is modelled on real retail behaviour and calibrated against a real published dataset.
"Then why not just use real data?"	Public retail datasets are wholesale-led, from 2009–2011, and have no store, city, promotion or stock fields. They cannot answer Indian retail questions. And with real data nobody knows the true answer, so model accuracy cannot be measured.
"How do we know it is any good?"	54 automated checks pass, and the output was calibrated against a real dataset — for example 19.4% of products generate 80% of revenue here, against 19.6% in real data.
"How long did it take?"	The generator runs in about 7 seconds once written. The work was in modelling the behaviour and validating it, not in producing volume.
"What did it cost?"	Nothing to license — it uses only the Python standard library. No tool purchase, no per-seat cost, no vendor lock-in.
"Can we delete it and regenerate?"	Yes. A fixed seed reproduces both files byte-for-byte.
"Can we use it for a customer-facing decision?"	Not on its own. It is right for building and testing. Anything commercially binding should be confirmed against live sales first.
"What are the biggest gaps?"	Returns, cost and margin, and customer demographics are not modelled. The forecasting panel also covers 150 of 1,200 products.
"Who built it?"	I built it — the generator, the data dictionary and the validation suite are all delivered alongside the data.

8. What is delivered

File	Size	Purpose
retail_pulse_sales.csv	45.4 MB	2,50,000 transaction line-items, 27 columns — customer insight and reporting
retail_pulse_demand_panel.csv	63.4 MB	7,80,000 rows, 15 columns, store × product × week — forecasting input
retail_pulse_data_dictionary.md	—	Every column defined, business rules, validation results and limitations
generate_retail_pulse.py	—	Regenerates both files from the seed
retail_pulse.py	—	Loads the data with types applied and re-checks the rules
verify_dataset.py , audit_quality.py	—	Re-runnable checks behind every figure in section 4

Retail Pulse AI — synthetic retail dataset. Figures computed directly from the delivered CSV files. Synthetic data for development and modelling; not a substitute for live transaction data in commercial decisions.